⏱️ Latency, Availability & Rate Limits

Week 4 • Day 4 • Technical Architecture & Constraints

TPM Academy

🎯 Day 4 Objectives

- Distinguish **SLI / SLO / SLA / error budget**
- Set the **three SLOs**: latency, availability, throughput / rate limit
- Walk the **latency budget** across your Container diagram — does the math fit?
- Design a **rate-limit policy** with explicit failure-mode behavior
- Name an **error-budget consequence** — the behavior change when we breach

🕑 Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Pin Container diagram; recap SLI/SLO/SLA
09:15 – 10:30	Block 1 + Activity 1	SLO Triage
10:45 – 12:00	Block 2 + Activity 2	Set your three SLOs
13:00 – 14:15	Block 3 + Activity 3	Latency-budget walk
14:30 – 16:00	Block 4 + Activity 4 + Wrap	Cross-review + AI sanity check

Block 1 — The Vocabulary

SLI / SLO / SLA / Error budget

📚 Get the Words Right

Term	Definition	Audience
SLI	Measurement: latency, availability, error rate	Operators / engineers
SLO	Target for the SLI	TPMs, eng leads
SLA	Contractual commitment with consequences	Customer-facing
Error budget	(1 − SLO) over the window	Eng leads, TPMs

Common error: "We commit to 99.99%" written casually = SLA, not SLO. Use SLO for internal targets; SLA only when contractually intended.

🎯 Three SLOs Are Usually Enough

Latency

"p ≤ ms % of the time over "

Anchor: when does the user give up?

Availability

"% of requests succeed over "

Anchor: cost of a minute of downtime?

Throughput / rate limit

"system handles rps per "

Anchor: what abuse / accident are we protecting against?

Each must have: a percentile, a window, and a defense.

⚠️ SLO Failure Modes

Failure	Example
No window	"p95 < 400ms" (over what?)
Average instead of percentile	"avg < 500ms" (hides bimodal distributions)
No defense	"99.9%" (why this number?)
Aspirational	"99.99%" without multi-region active-active
User-mismatched	"p95 < 5s" for a typing-feedback feature

👥 Activity 1 — SLO Triage

Triads • 35 minutes

8 SLO statements. Identify failures. Rewrite the 5 worst.

- 15 min: triage
- 15 min: rewrite (percentile + window + defense)
- 5 min: identify the most subtle failure

⏱️ 15 + 15 + 5

Block 2 — Set Your Three SLOs

Latency / Availability / Rate limit

📎 Error Budget — the Decision Tool

If availability SLO is **99.9%**, error budget is **0.1%** — about **43 minutes** of downtime per 30 days.

- **Inside the budget** → ship features (more risk-tolerant)
- **Near or past the budget** → prioritize reliability (more risk-averse)

Converts SLO from abstract to load-bearing: a team that respects its error budget is doing SRE, even if not labeled as such.

📝 Worked SLOs (FieldPulse reconcile)

```
### SLO — Latency
**Measurement:** Reconcile modal open time, mobile client.
**Target:** p95 ≤ 1000ms, 99% over 30 days.
**Defense:** Dispatchers tap reconcile 30+ times/shift; > 1s
crosses the "I'd rather use paper" threshold (Wk 1
interview Maria, Tier Sheet op-signal #2).
**Verification:** Synthetic transaction in Datadog.

<h3 id="slo--availability">SLO — Availability</h3>
<p><strong>Measurement:</strong> Successful (non-5xx) reconcile submissions.
<strong>Target:</strong> 99.5% over 30 days.
<strong>Error budget:</strong> ~3.6 hrs/month.
<strong>Defense:</strong> Dispatch volume averages 200 reconciles/day; 30-min
outage during peak shift = ~50 affected dispatchers.
<strong>Verification:</strong> Pingdom + 5xx alerting.
<strong>Budget consequence:</strong> If 7-day rolling burns >50% of budget,
we freeze new feature merges and prioritize reliability.</p>
<h3 id="rate-limit-policy">Rate-limit policy</h3>
```

```
<strong>Per-user:</strong> 10 reconcile-submits/sec, burst 30.
<strong>Per-tenant:</strong> 100 reconcile-submits/sec.
<strong>Failure mode:</strong> HTTP 429 with Retry-After header; client
implements exponential backoff with jitter.
<strong>Defense:</strong> Protects against runaway script bug seen in the
2024 dispatcher-tool incident.
<strong>Verification:</strong> Load test before launch; rate-limit cap dashboard.
```

👥 Activity 2 — Set Your Three SLOs

Triads • 40 minutes

- 15 min: latency SLO (percentile + window + defense)
- 10 min: availability SLO + error-budget translation
- 15 min: rate-limit policy with failure-mode behavior

⏱️ 15 + 10 + 15

Block 3 — Latency Budget Walk

Does the math fit your Container diagram?

📊 Per-Hop Latency Reference (rough)

Hop type	Typical range
Network round-trip (LAN → Internet)	20–80ms
Database read (indexed)	5–30ms
Database write (single-row)	10–50ms
External API call (sync)	50–500ms (varies)
Async publish (Kafka/SQS local)	1–10ms
JWT validation	1–5ms
TLS handshake (cold)	50–200ms

These are starting points. Use real platform numbers if available.

📊 The Latency Math

```
SLO target: p95 ≤ 1000ms

<p>Path: Mobile → API GW → Reconcile → Postgres read
→ Tickets module
→ Postgres write
→ Audit publish
→ response back</p>
```

```
Estimated:
Mobile→API GW round trip:    80ms
API GW → Reconcile:           20ms
Postgres read:
  30ms
Tickets module call (sync):  100ms ← dominates
Postgres write:
 40ms
Audit publish (async):        10ms
Response back:
  80ms
─────
~360ms median
p95 likely ~700–900ms ✓
```

If math doesn't fit: tighter SLO / fewer hops / faster hops / wider percentile.

👥 Activity 3 — Latency-Budget Walk

Triads • 40 minutes

Annotate your Container diagram with per-hop latency. Sum the path. Decide what gives if it doesn't fit.

- 15 min: annotate per-hop latency
- 10 min: sum the path
- 10 min: if doesn't fit, decide what gives
- 5 min: flag question for engineering

⏱️ 15 + 10 + 10 + 5

Block 4 — Cross-Review + AI Sanity Check

Three questions; one prompt; honest revision

🔎 The Three Cross-Review Questions

1. **Latency target realistic** given platform precedent?
2. **Availability target realistic** given the team's on-call reality?
3. **Rate-limit policy** protects against a real abuse / accident scenario?

> **Author triad listens, captures, decides.** Same Week-3 review discipline.

AI Sanity Check

```
Role: Senior reliability engineer reviewing a feature's SLO sheet.
Context: <paste slo="" sheet="" +="" latency="" walk="" rate-limit="" policy="">
Architecture stance: <paste tcd="" §1="">
Task: Identify the top 3 reasons the SLOs as written might
not be sustainable.
Constraints:
- Use only the information provided; do not invent platform details
- Be specific: which SLO, which scenario, which condition
- Suggest a more realistic alternative for each
Format: Numbered list — Issue / Scenario / Suggested target.</paste></paste>
```

👥 Activity 4 — Cross-Review + AI

Triad pairs • 45 minutes • Wrap

- 20 min: cross-review (3 questions)
- 15 min: AI sanity-check + adopt/defer/reject
- 10 min: revise §4 + provenance note

⏱️ 20 + 15 + 10 · 15-min wrap

🎉 Day 4 Wrap

What we built

- SLI / SLO / SLA / error-budget vocabulary
- Three SLOs (latency / availability / rate)
- Latency-budget walk through Container diagram
- Error-budget consequence stated
- TCD §4 drafted

What opens tomorrow

- **Technical trade-offs & constraints**
- Friday: assemble TCD §§5–6
- Stakeholder + sign-off matrix
- Cross-triad review of the full TCD

Bring to Day 5: Your TCD §§1–4. Tomorrow integrates everything into the final TCD.

💬 Questions & Reflection

Which of your SLOs would you fold under pressure from a stakeholder asking for "just a bit more"?